# Domain and range

**1**

**a** Any value of $x$.

**b** Only non-negative values of $y$.

**2**

**a** $0$

**b** There is no greatest value.

**c** $y \geq 0$

**d** All real $x$

**3**

**a** $0$

**b** There is no greatest value.

**c** $y \geq 0$

**d** All real $x$

**4** Example answer: $f(x) = |x| + 2$

**5**

**a** $[0, 6]$

**b** $(3, 6]$

**6**

**a**

**i** All real $x$ **ii** $y \leq 0$

**b**

**i** All real $x$ **ii** $y \geq -1$

**c**

**i** All real $x$ **ii** $y \leq 4$

**d**

**i** All real $x$ **ii** $y \leq 2$

**7**

**a**

**i** $[-5, -2]$ **ii** $[0, 6]$

**b**

**i** $[-2, 3)$ **ii** $[-9, 0]$

**c**

**i** $(-2, 1)$ **ii** $[0, 6)$

**d**

**i** $[2, 4)$ **ii** $(0, 4]$

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## Graphs of absolute value functions

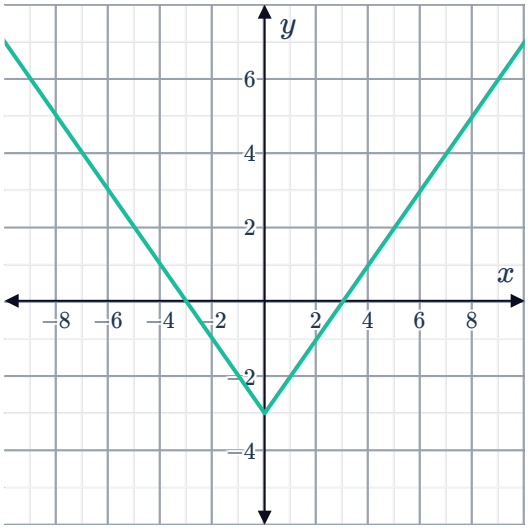
**8**

**a**

**i** All real $x$

**ii** $y \geq -3$

**iii**

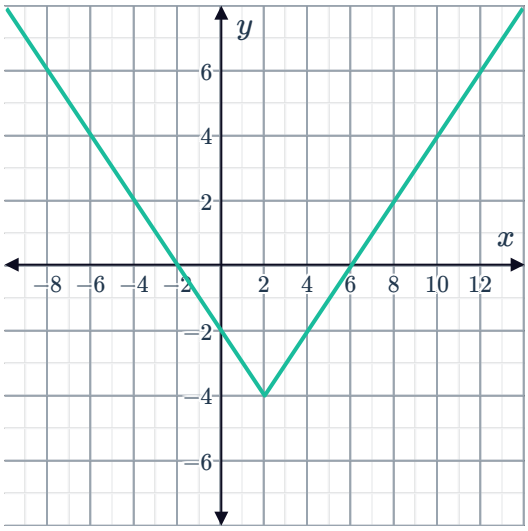



**b**

**i** All real $x$

**ii** $y \geq -4$

**iii**



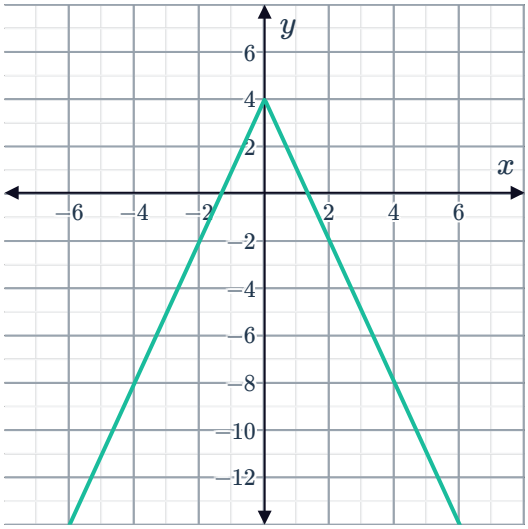

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**c**

**i** All real $x$

**ii** $y \leq 4$

**iii**

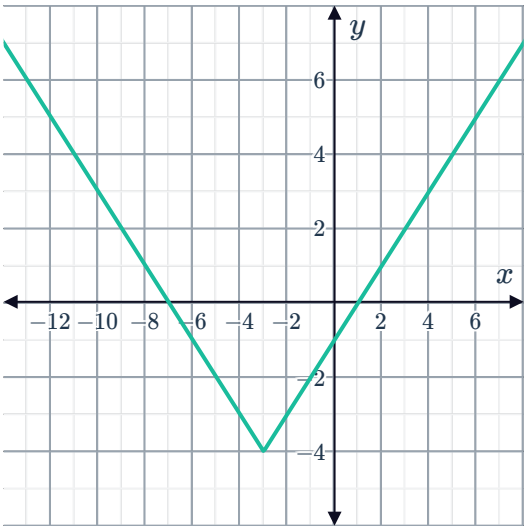



**9**

**a**

**i** $y \geq -4$

**ii**




**iii** $(-3, -4)$

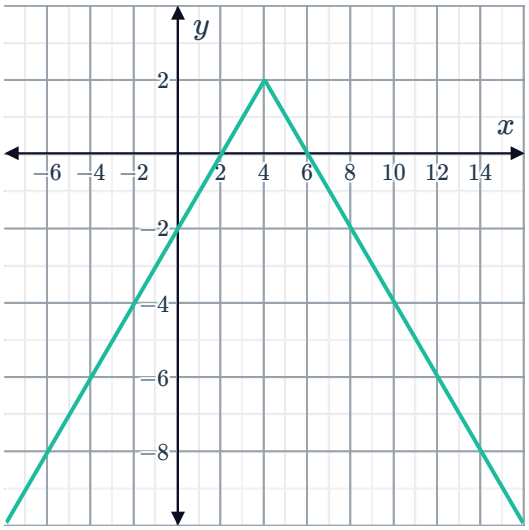
**iv** $-4$

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**b**

**i** $y \leq 2$

**ii**




**iii** $(4, 2)$

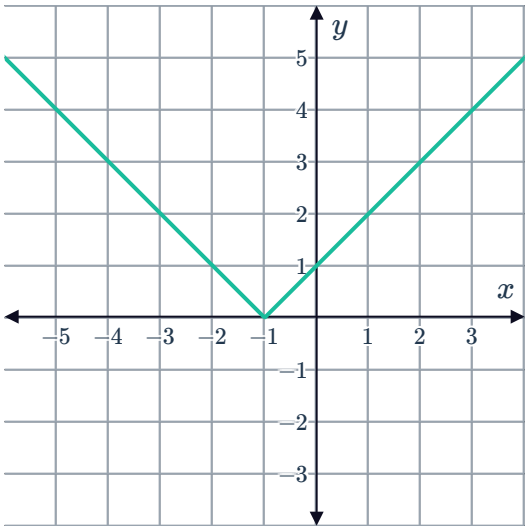
**iv** $2$

**10**

**a**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $1$ | $0$ | $1$ | $2$ | $3$ |

**b**




**c** $x = -1$

**d** $(-1, 0)$

**e**

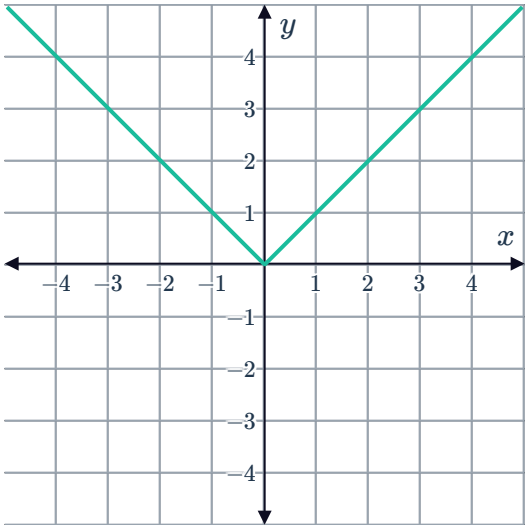
| | Equation | Slope |
|---|---|---|
| $x < -1$ | $y = -x - 1$ | $-1$ |
| $x > -1$ | $y = x + 1$ | $1$ |

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**11**

**a**

**i**



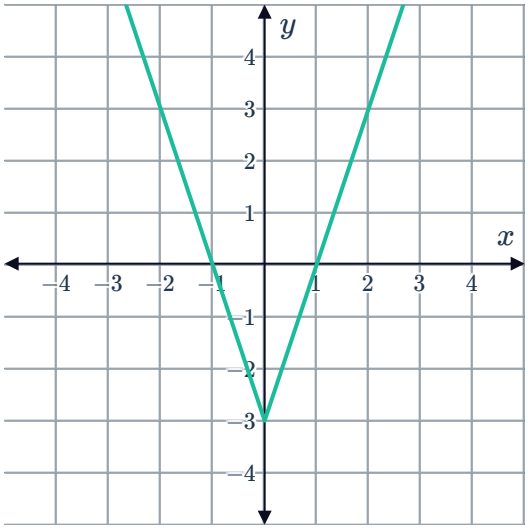

**ii** $(0, 0)$

**iii** $-\infty < x < \infty$

**iv** $y \geq 0$

**b**

**i**




**ii** $(0, -3)$

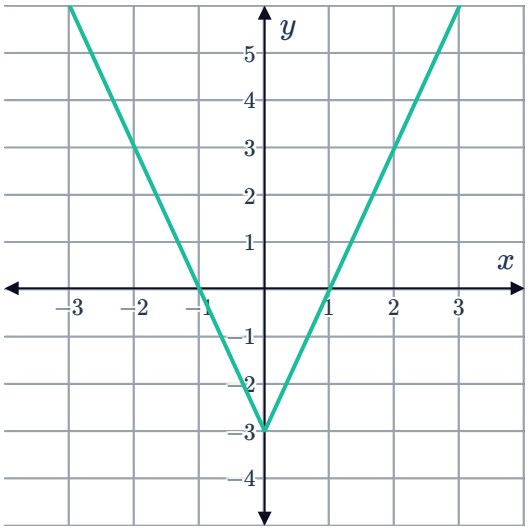
**iii** $-\infty < x < \infty$

**iv** $y \geq -3$

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**12**

**a**




**b** $x = 0$

**c** $(0, -3)$

**d** $x = -1, 1$

**e** $y = -3$

**f** $-3$

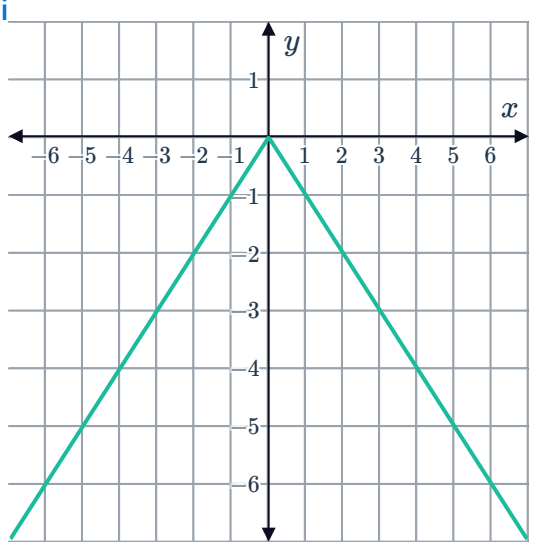
**g** $3$

**13**

**a**

**i** Downwards **ii** $(0, 0)$

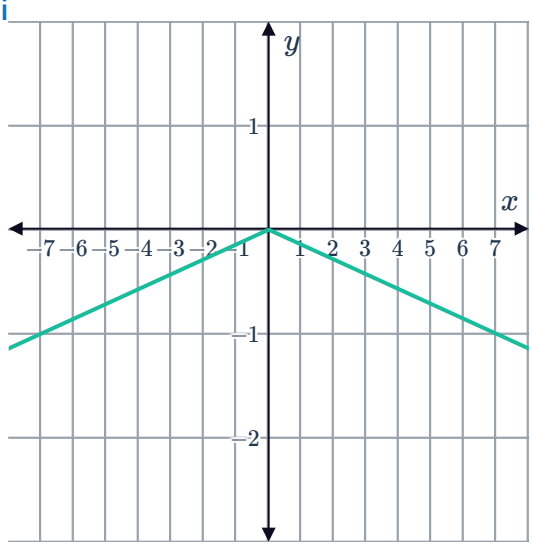
**iii**




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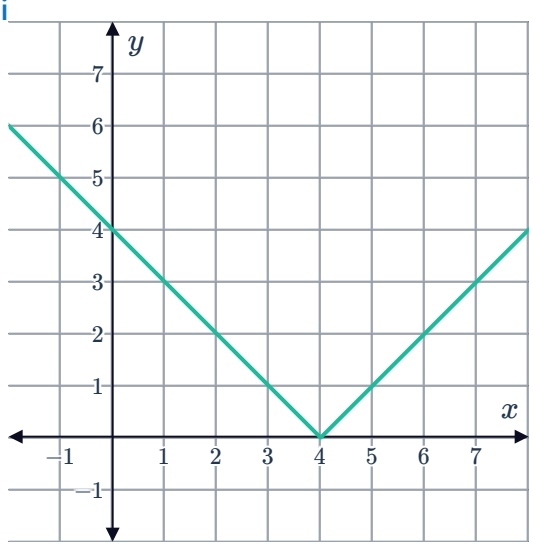
**b**

**i** Downwards **ii** $(0, 0)$ **iii**



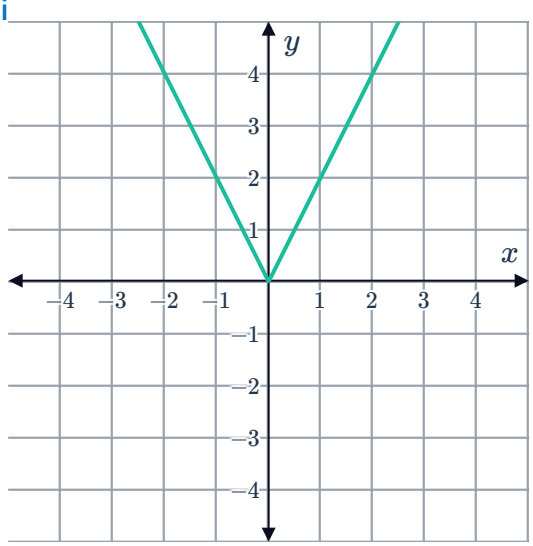

**c**

**i** Upwards **ii** $(4, 0)$ **iii**




**d**

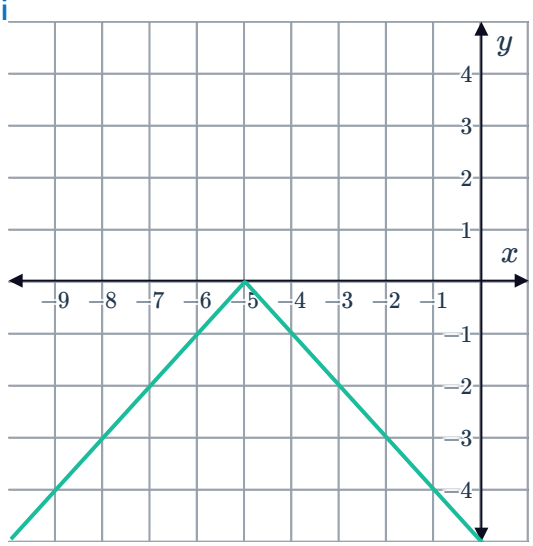
**i** Upwards **ii** $(0, 0)$ **iii**




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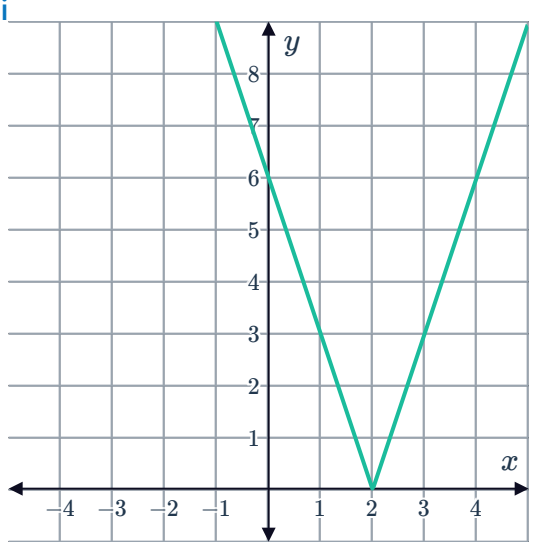
**e**

i Downwards ii $(-5, 0)$ iii



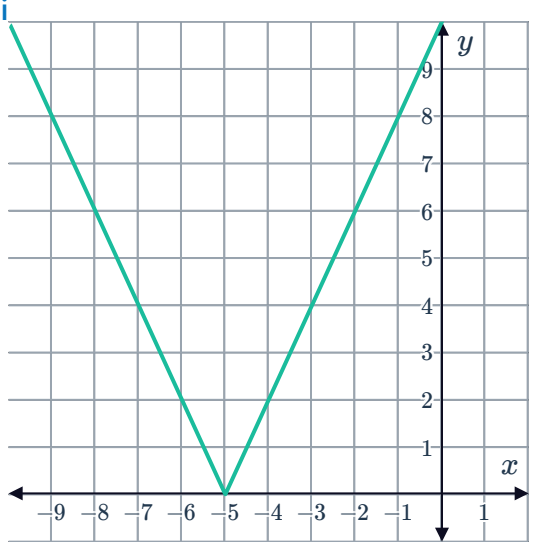

**f**

i Upwards ii $(2, 0)$ iii




**g**

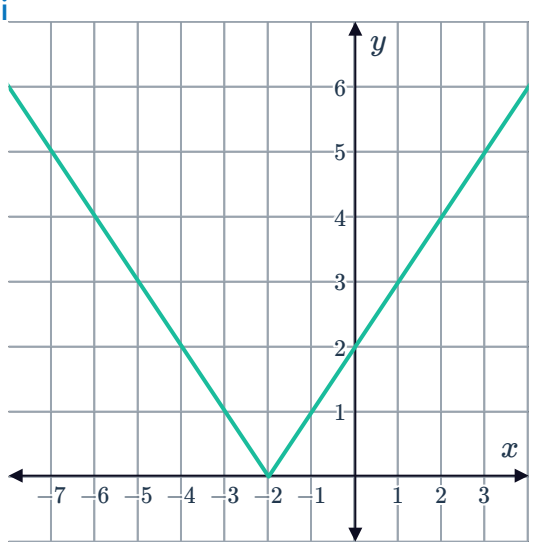
i Upwards ii $(-5, 0)$ iii




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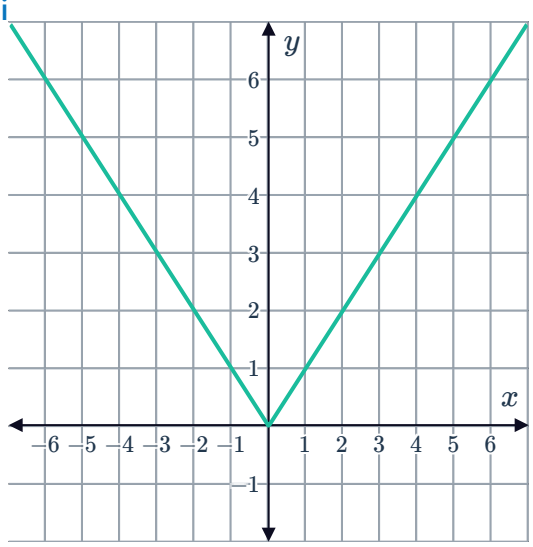
**h**

**i** Upwards **ii** $(-2, 0)$ **iii**




**i**

**i** Upwards **ii** $(0, 0)$ **iii**




**14**

**a** Upwards **b** $(-1, 0)$ **c** $x = -1$ **d**

**i** No **ii** No

**iii** Yes

**15**

**a** Upwards **b** $(3, 0)$ **c** $x = 3$ **d**

**i** No **ii** No

**iii** Yes **iv** No

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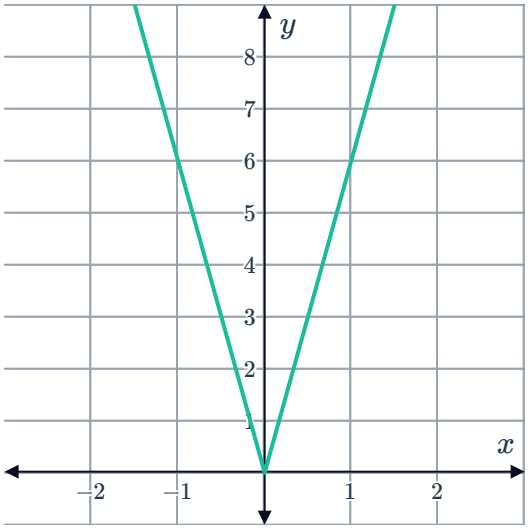
**16**

**a** Upwards

**b** $x = 0$

**c** $(0, 0)$

**d**




**e**

**i** No **ii** No **iii** Yes **iv** Yes

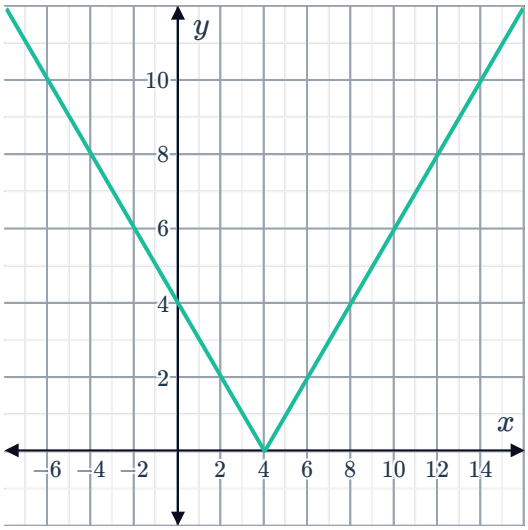
**17**

**a** Upwards

**b** $x = 4$

**c** $(4, 0)$

**d**




**e**

**i** Yes **ii** No **iii** No **iv** Yes

**18**

**a** $3$

**b** $f(x) = |3(x - 4)|$

**c** $(-\infty, \infty)$

**d** $[0, \infty)$

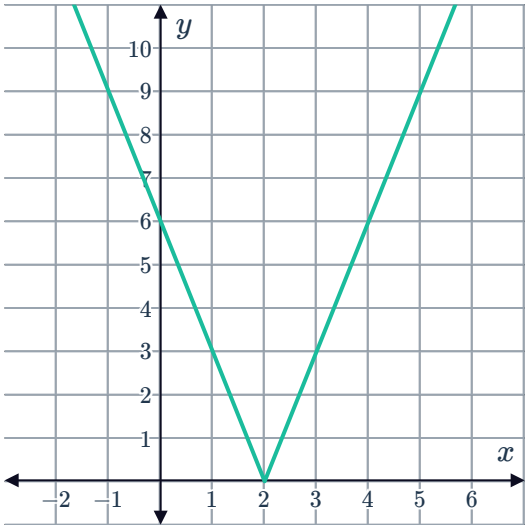
**e**

**i** Yes **ii** No **iii** No

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**19**

**a**

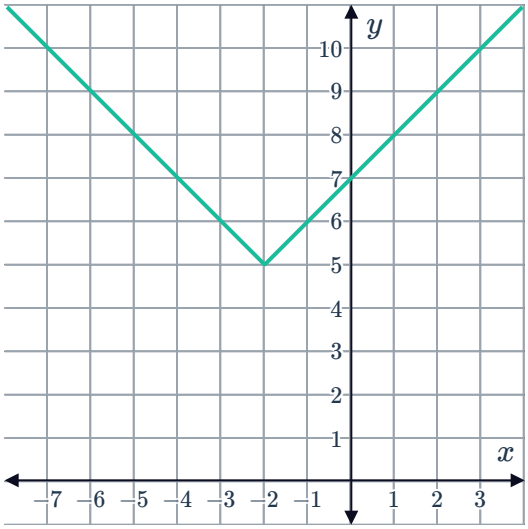



**b** $(2, 0)$

**20**

**a**

**i**

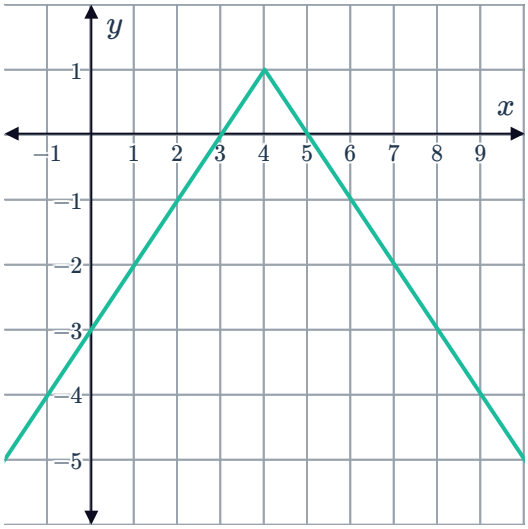



**ii** $(-2, 5)$

**iii** $5$

**b**

**i**



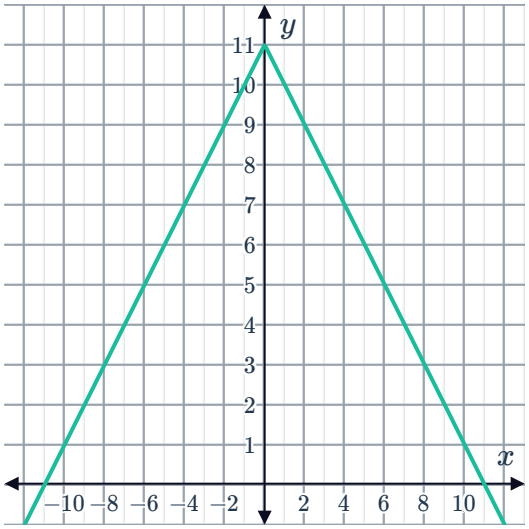

**ii** $(4, 1)$

**iii** $1$

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**21**

**a**



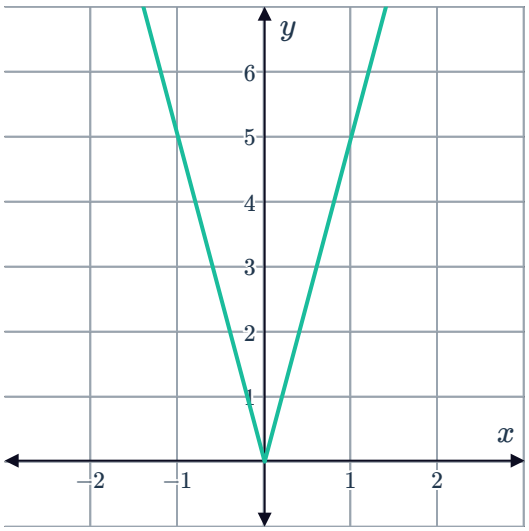

**b** 11

**c** $(-\infty, 0)$

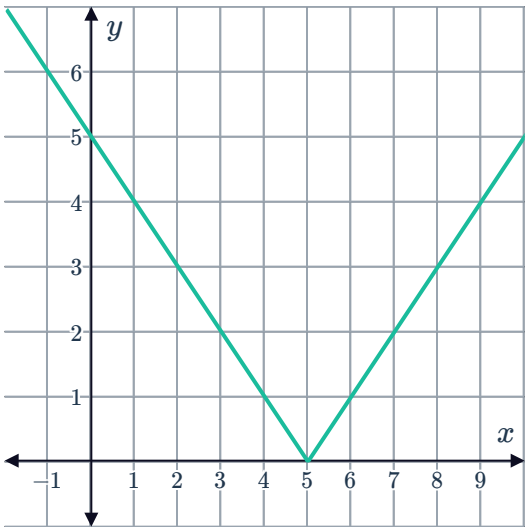
**d** $(0, \infty)$

**22**

**a**

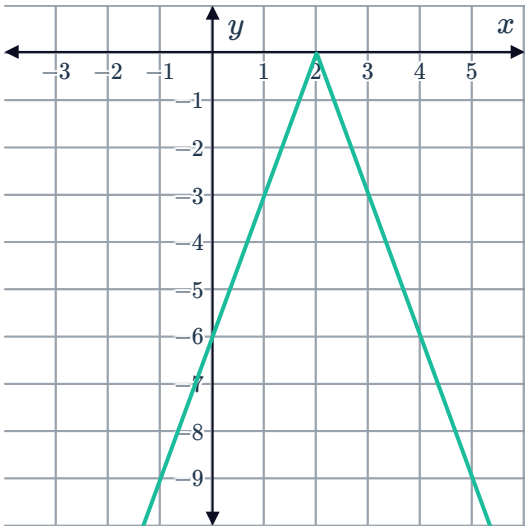



**b**

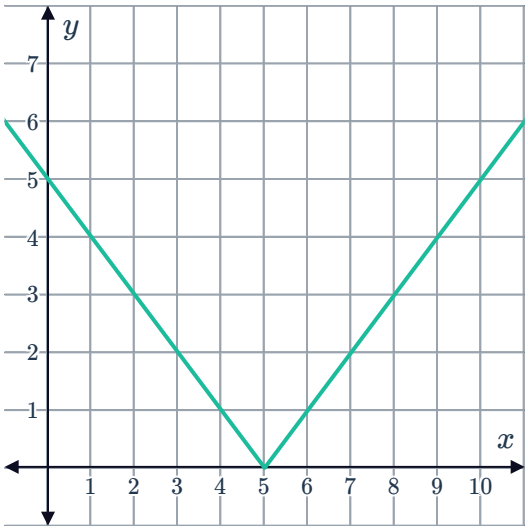



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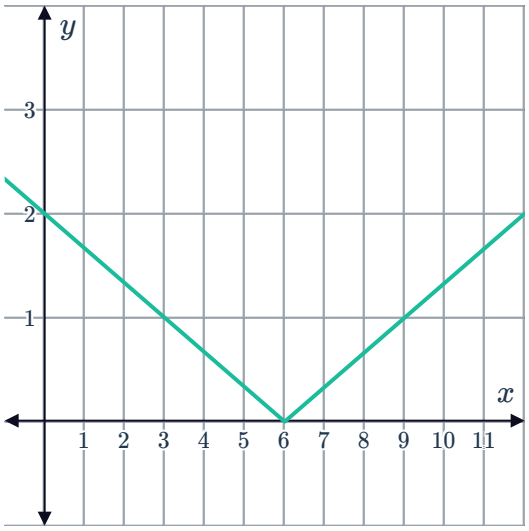
c



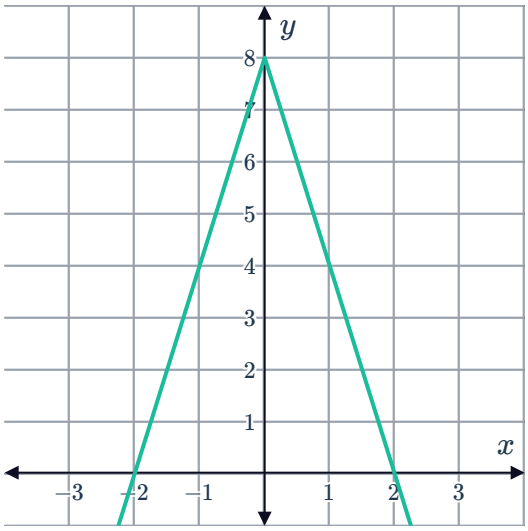

d



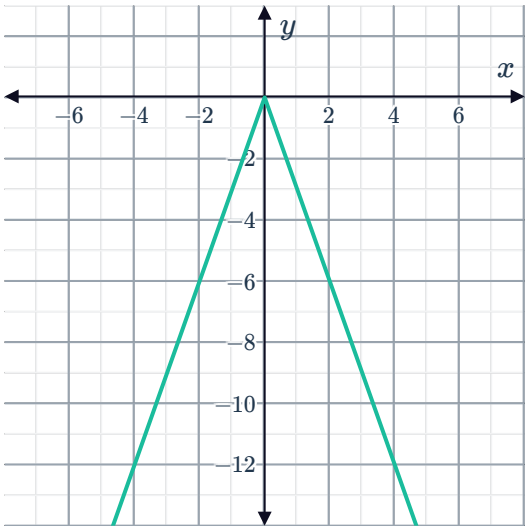

e



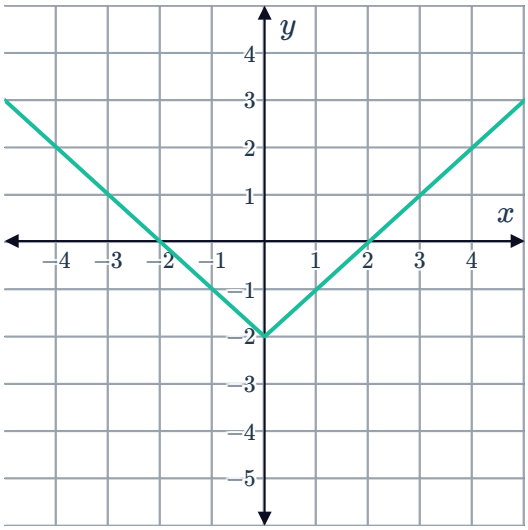

f




g

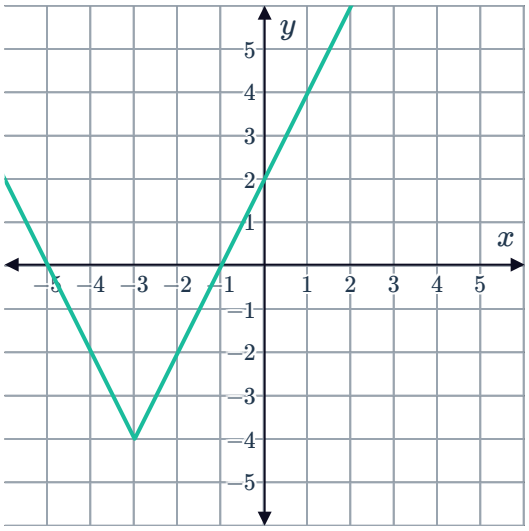



h

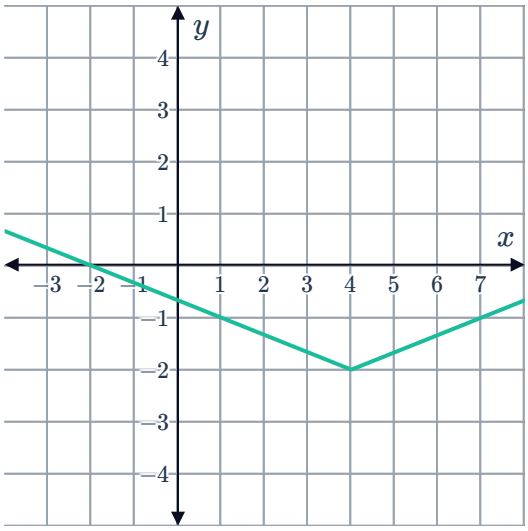



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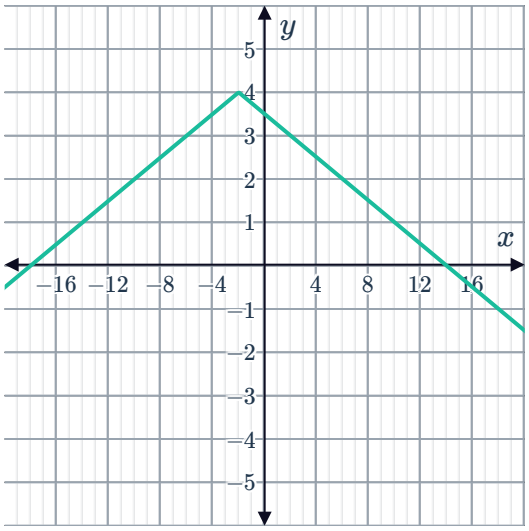
i




j



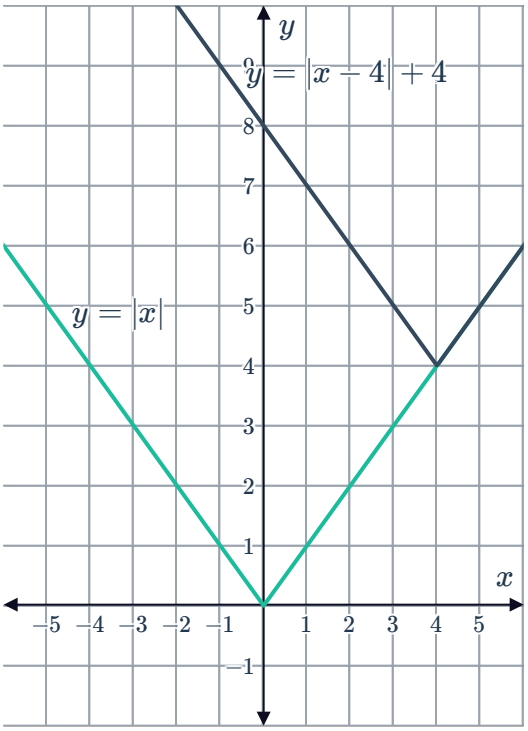

k




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## Transformations

**23**




**24** $f(x) = |x + 2|$

**25**

**a** $y = |x + 5| - 3$

**b** $y = -4\,|x|$

**c** $y = -\dfrac{|x + 3|}{4} + 3$

**26**

**a** Horizontal translation

**b** Yes

**c** Yes

**27** Translation of $4$ units down

**28** Vertical stretching by a factor of $3$.

**29**

**a** Reflection about the $x$-axis and translation of $4$ units up.

**b** $f(x) = 4 - |x|$

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**30** **a** $3$

**b** $-3$

**c** $(0, 0)$

**d** $(2, -3)$

**e** Vertical stretching by a factor of $3$, translation of $2$ units to the right and $3$ units down

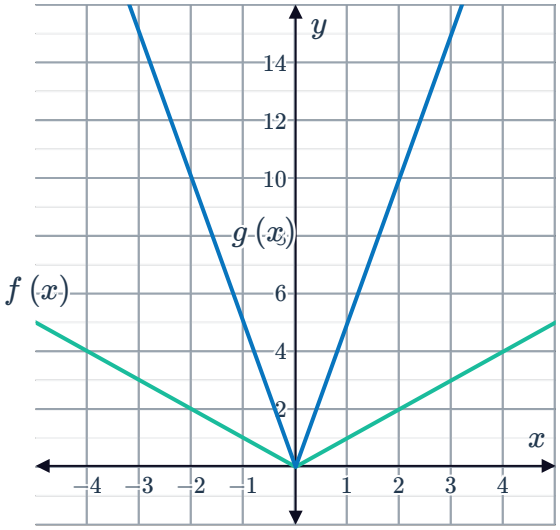
**f** $f(x) = 3\,|x - 2| - 3$

**31**

**a**

| $x$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|---|
| $y = f\,(x)$ | $3$ | $2$ | $1$ | $0$ | $1$ | $2$ | $3$ |
| $g\,(x) = 5f\,(x)$ | $15$ | $10$ | $5$ | $0$ | $5$ | $10$ | $15$ |
| $h\,(x) = -2f\,(x)$ | $-6$ | $-4$ | $-2$ | $0$ | $-2$ | $-4$ | $-6$ |

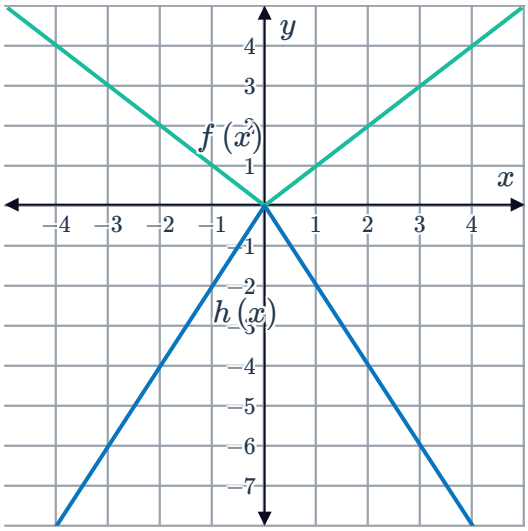
**b**




**c** vertically stretched by a factor of $5$

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**d**




**e** Vertically stretched by a factor of $2$ and reflected about the $x$-axis

## Applications

**32**

**a** $80\ \text{mi}$ **b** $8$ minutes **c** Slower

**33**

**a** $0 \leq x \leq 60$

**b** $120\ \text{km}$

**c** $30$ minutes

**d** Faster, because the lines are steeper than in the first graph.

**34**

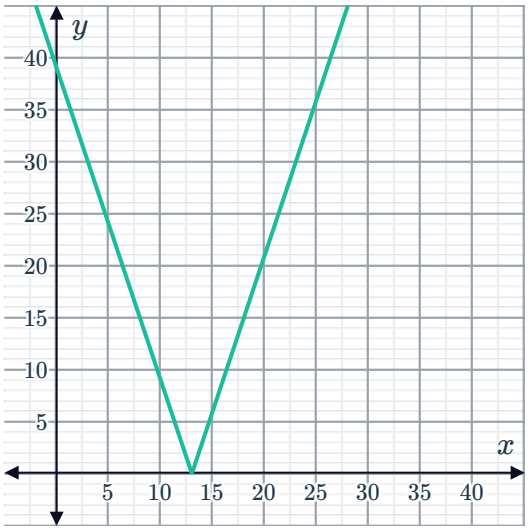
**a** $700\ \text{mi/h}$ **b** $t = \left| \frac{3}{35}x - 90 \right|$ **c** $90$ minutes **d** $2.5$ hours

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**35**

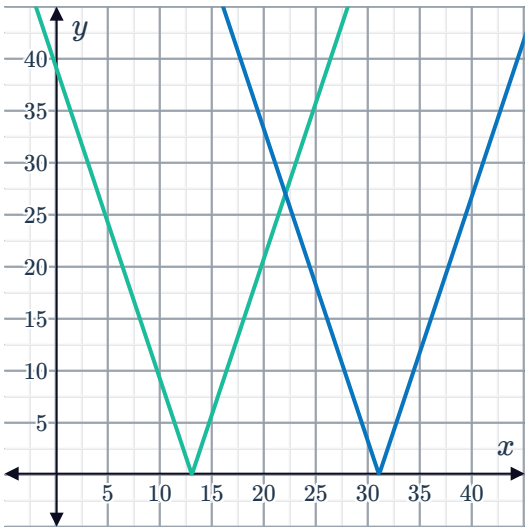
**a** $y = 3\,|x - 13|$

**b**




**c** $y = 3\,|x - 31|$

**d**




**e**

**i** Yes **ii** No **iii** Yes **iv** No

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